USN | | | | | | | | | | [23ESME013]

FIRST SEMESTER B.TECH EXAMINATION, JANUARY - 2024

BASICS OF CIVIL AND MECHANICAL ENGINEERING

Time: 03 HOURS **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer all questions.

		PART - A	**05 X 02=10**		
1.	**a.**	List the properties of ferrous materials.	L2	CO1	**02**
	b.	What is an impulse turbine?	L2	CO2	**02**
	c.	Classify the types of boilers with suitable examples.	L2	CO3	**02**
	d.	List different disciplines of civil engineering.	L2	CO4	**02**
	e.	Classify the types of roads according to materials.	L2	CO5	**02**
		PART – B	**06×05=30**		
2.	**a.**	Differentiate between the impulse and reaction turbines.	L2	CO1	**05**
		OR			
	b.	Explain the working of thermal power plant with neat sketch.	L2	CO	**05**
3.	**a.**	Is hydel power plant can be used for base load? Also explain its working principle.	L2	CO2	**05**
		OR			
	b.	Explain any five properties of metals.	L2	CO2	**05**
4.	**a.**	Write the classification of IC Engines.	L2	CO3	**05**
		OR			
	b.	What are the advantages and disadvantages of thermal power plants for electricity generation?	L2	CO3	**05**
5.	**a.**	Discuss the emerging areas in civil engineering.	L2	CO4	**05**
		OR			
	b.	Explain in detail about water security.	L3	CO4	**05**
6.	**a.**	Discuss in detail importance of any two building materials.	L3	CO5	**05**
		OR			
	b.	Explain in detail Rain water harvesting.	L3	CO5	**05**
7.	**a.**	Explain different mode of heat transfer.	L2	CO3	**05**
		OR			
	b.	What are ceramic materials? List its properties.	L2	CO3	**05**
		PART – C	**06×10=60**		
8.	**a.**	Explain various types of hybrid vehicle configurations.	L2	CO1	**10**
		OR			
	b.	Explain the key components of a hybrid electric car.	L2	CO1	**10**

9.	**a.**	Explain the working of a vapor compression refrigeration system.	L2	CO2	**10**
		OR			
	b.	Explain the working principle of vapor absorption refrigeration system.	L2	CO2	**10**
10.	**a.**	Explain the working of the Cochran boiler with a neat sketch.	L2	CO3	**10**
		OR			
	b.	Explain following. **i.** Refrigeration effect **ii.** Ton of refrigeration **iii.** Coefficient of performance	L2	CO3	**10**
11.	**a.**	Explain in detail surveying, geotechnical engineering, and structural engineering.	L3	CO4	**10**
		OR			
	b.	Explain the advantages and disadvantages of automation and robotics in construction industries.	L3	CO4	**10**
12.	**a.**	Explain in detail environmental issues concerned to building materials.	L3	CO5	**10**
		OR			
	b.	Explain the global warming and its impact by the construction industry.	L3	CO5	**10**
13.	**a.**	Explain the difference between TIG and MIG welding, explain anyone with a neat sketch.	L2	CO2	**10**
		OR			
	b.	Explain the factors that affect the efficiency of a refrigeration system.	L2	CO3	**10**